

SUMMARY

Physical AI and Robotics Engineer in Research & Development at the Industrial Autonomy & Engineering Lab Powered by NVIDIA, specializing in robot control, motion planning, and autonomous systems. Focused on Physical AI and advanced robotics R&D with experience in simulation-to-real-world deployment. Skilled in AI model deployment, multi-sensor fusion, and computer vision for perception-driven autonomy, with strong grounding in sensor calibration, localization, and mapping fundamentals underlying SLAM and multi-sensor perception pipelines. Experienced in developing scalable multi-robot systems using industrial-grade software, firmware, sensors, and embedded platforms for robust real-time performance in complex environments.

TECHNICAL SKILLS

Languages: Python, C++, Embedded C, Java

Robotics: ROS1/ROS2, Nav2, SLAM, RTAB-Map, VSLAM, Visual-Inertial Odometry (VIO), Localization, Global & Local Path Planning, A*, RRT, DWB, TEB, Motion Planning, MoveIt, Inverse Kinematics, cuRobo, Robotic Manipulator Control, Pick and Place, Obstacle Avoidance, Sensor Fusion & Calibration (LiDAR, IMU, Camera), Point Cloud Processing, GraspNet, AMR, Cobots System, PID Tuning, Feedback Control Systems

AI/ML & Perception: Deep Learning, Machine Learning, PyTorch, TensorFlow, OpenCV, TensorRT, YOLO, Object Detection, Image Segmentation, Classification, Depth Estimation, Reinforcement Learning, SAC, PPO, Vision-Language-Action (VLA), Embodied AI

Simulation & Modelling: NVIDIA Isaac Sim, Gazebo, MuJoCo, URDF/Xacro/SDF/USD, AutoCAD, Digital Twin, Synthetic Data Generation, Sim-to-Real Transfer

NVIDIA Stack: Jetson Orin, Jetson Nano/NX, CUDA, NVIDIA Omniverse, Isaac ROS, Isaac Lab, Isaac GR00T, Cosmos Transfer, NVIDIA Drive, Edge AI, Edge Deployment, On-Device LLM Inference (Ollama, Qwen)

Embedded & Tools: STM32, Raspberry Pi, ESP32, RTOS, Git, GitHub, Linux/Ubuntu

EXPERIENCE

Robotics Engineer — Tata Consultancy Services *Industrial Autonomy and Engineering Lab Powered by NVIDIA* / Bengaluru | Apr 2025 – Present

- Architected an end-to-end AMR navigation stack using the Navigation Framework, combining Karto SLAM, AMCL localization, IMU and wheel-odometry sensor fusion, and A*, DWB and TEB planners with a custom robot control SDK and firmware matched to platform drive kinematics.
- Built multi-sensor fusion pipelines (LiDAR, IMU, camera) for 2D/3D mapping with tuned costmap layers, implementing Google Ceres Solver for sparse pose-graph optimization and tuned SLAM localization, achieved 10mm loop closure error with drift-free mapping across complex environments.
- Built real-time dynamic obstacle avoidance using layered costmaps (static, obstacle, inflation) fused with LiDAR and depth camera perception, extending reaction distance via Kalman-filter-based obstacle velocity prediction to anticipate dynamic obstacle trajectories. Reduced perception-to-planning latency through GPU-accelerated point cloud voxelization, sustaining local planner control loop above 25Hz.
- Developed a GPU-accelerated intelligent perception and manipulation pipeline: integrated NVIDIA Cosmos Reasoning for scene-level perception reasoning, quantized via TensorRT (INT8 post-training quantization) to reduce inference latency for real-time deployment; combined with cuRobo for GPU-accelerated inverse kinematics and GraspNet for point-cloud-based grasp pose detection, driving the pipeline toward fast, low-latency perception reasoning and action execution.
- Solved 3D map generation from monocular camera input alone using a ViT-based monocular depth estimation model, benchmarking AbsRel and RMSE accuracy against dedicated depth-sensor ground truth across near/mid/far range buckets, matched depth-sensor-equivalent accuracy at navigation-relevant ranges, validating deployment without additional depth hardware.
- Delivered manipulator control and perception-driven pick-and-place using depth models, reinforcement learning, and inverse kinematics, deployed AI-enabled cobots from simulation to production with AMR cobot arm coordination.
- Japan-based electronics manufacturer:** Built the plant's complete digital asset set (conveyors, multiple robotic arms) and a real-time synchronized digital twin in Isaac Sim connected to the physical robotic arm, using AWS IoT Core to stream real-time machine data to the cloud.
- Japan-based robotics manufacturer:** Built a point-cloud-to-simulation pipeline fusing LiDAR and monocular imagery into calibrated 3D Isaac Sim assets, then trained an AMR for autonomous navigation in the reconstructed environment.

AWARDS

- Star of the Month, TCS:** Built a real-time, cloud-connected system linking a 6-DOF manipulator-equipped AMR with an LLM-based speech-to-action pipeline for cross-border remote operation, streaming live telemetry and action feedback to a synchronized digital twin in NVIDIA Omniverse.
- Innovation Spark Award, TCS:** Delivered an industrial-grade AMR end to end – mechanical design, control firmware using industrial-grade equipment, custom SDK, AI perception, dense mapping, complex obstacle avoidance, and AI deployment.

PERSONAL PROJECTS

- Digital Twin in NVIDIA Isaac Sim:** Created a digital asset of a real-world environment (URDF/Xacro/SDF/USD) in Isaac Sim and connected the physical robot to it in real time using the DDS API, generated synthetic training data and applied sim-to-real transfer for hardware deployment.
- Vision-Based Autonomous Navigation — Physical Mecanum Robot:** Built a real-world holonomic robot navigating autonomously using a vision-based reactive reinforcement learning policy, leveraging OpenCV and YOLOv8 object detection with multi-sensor fusion for obstacle-aware path decisions.
- Dynamic Obstacle Avoidance Using Potential Field & LiDAR:** Real-time reactive obstacle avoidance for a differential-drive robot in Gazebo.
- Deep Reinforcement Learning (SAC) for Robot Navigation:** Trained a mobile robot navigation policy using Soft Actor-Critic (SAC) in ROS2 and Gazebo, accelerating training via CUDA-based GPU parallelization, and optimized the trained policy for real-time inference using TensorRT.
- SLAM & Autonomous Navigation (ROS2, Gazebo Harmonic):** Built a custom robot performing LiDAR-based SLAM and multi-room environment mapping, using A* global and DWB local path planners for navigation, tuning velocity limits to reduce collision rate and maintain clearance at higher speeds.
- Voice-Controlled 6-DOF Pick-and-Place:** Built a 6-DOF manipulator system with vision-based object perception and on-device LLM (Ollama/Qwen) command interpretation, translating natural-language voice instructions into executed pick-and-place actions.

EDUCATION

B.Tech, Electrical Engineering — University of Science, Technology and Agriculture, CU, Kolkata *Sep 2021 – May 2024*

B.Sc, Physics (Honours) — Asutosh College, University of Calcutta, Kolkata

Jun 2018 – Aug 2021